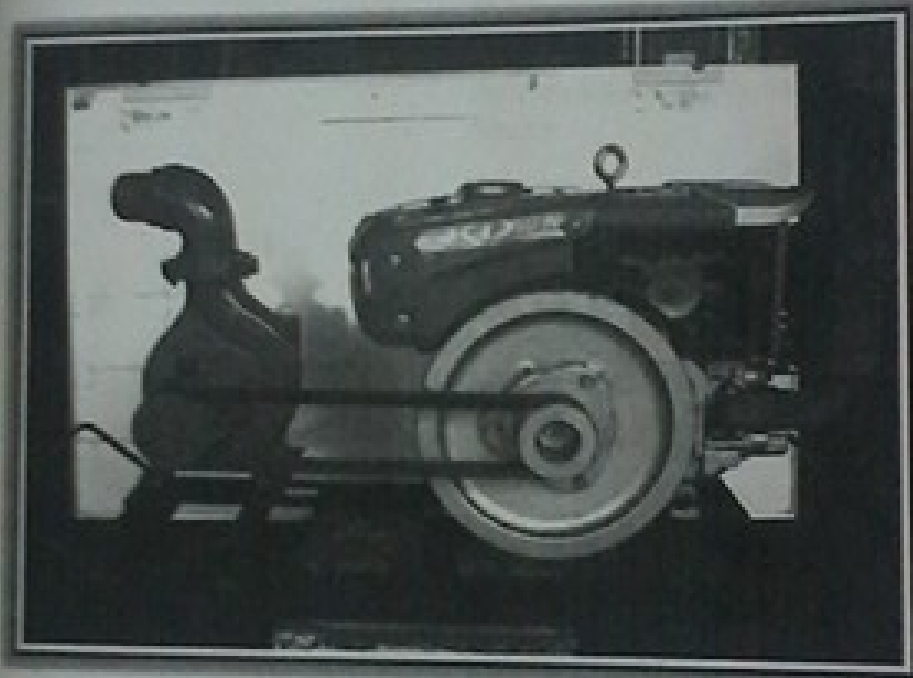




AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2014 - 39



**KATO KC30 3x3 Centrifugal Pump coupled to
KATO KD80N PH Diesel Engine**

KATO KC30 3"x3" Centrifugal Pump coupled to KATO KD80N PH Diesel Engine

Test Report No.	:	2014 - 39
Date of Test	:	April 1, 2014
Test Valid Until	:	April 1, 2019
Brand	:	KATO
Model	:	KC30 3" x 3"
Serial No.	:	No Data
Type and Size	:	Single-end suction, 75 x 75 mm, Self-Priming Centrifugal pumpset
Manufacturer	:	KATO INTERNATIONAL, INC. Valenzuela City
Test Requested by	:	TRAMAT MERCANTILE, INC. 747-749 Sabino Padilla, St., Sta. Cruz, Manila
Machine Classification	:	Commercial

SUMMARY

The KATO KC30 3"x3" Self-priming Centrifugal pump coupled to KATO KD80N PH Diesel Engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with 102 mm (4.0 inches) and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 25.0 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2421 and 2494 rpm, respectively, with Engine Fuel Consumption Rate of 1.26 L/h. The Maximum Discharge Capacity of 21.3 L/s occurred at 11.0 m Total Head with Fuel Consumption Rate of 2.27 L/h. The shaft speed of the pump and engine at this point were 2199 and 2294 rpm, respectively. The Maximum System Efficiency of 15.0% occurred at 16.1 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.57 m. The Net Positive Suction Head Available at this point was 2.40 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in Figure 1.

The test was conducted on April 1, 2014 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. KATO KC30 3"x3" Centrifugal Pump coupled to KATO KD80N PH Diesel Engine on the test set-up

METHOD OF TEST

The pump was tested following the "Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery - Centrifugal, Mixed-Flow, Axial-Flow Water Pumps - Methods of Test".

DESCRIPTION OF THE PUMP

The KATO KC30 Self-priming Centrifugal Pump shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump close-coupled to a 5.97 kW (8.0 Hp) KATO KD80 PH Diesel Engine. It was mounted on a steel base. The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a enclosed type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron dual volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. Located on top of the pump was the discharge outlet. The stuffing box used a packing type seal.

The primemover, a 5.97 kW KATO KD80N PH Diesel Engine was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was gravity type with the fuel tank located on top of the engine. It utilized an oil bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by water circulating from cylinder jacket to the radiator using thermo-siphon principle. A fan driven by the pulley on the flywheel was used to cool the warm water in the radiator. It was started using manual cranking lever aided by a manually actuated decompression lever. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.



Figure 2. KATO KC30 3"x3" Centrifugal Pump coupled to
KATO KD80N PH Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KATO KCM 3"X3" Centrifugal Pump coupled to KATO KEMIN PH Diesel Engine

1	Brand	:	KATO
	Model	:	KCM 3"X3"
	Serial No.	:	No Data
	Made	:	KATO INTERNATIONAL, INC. Valencia City
2	Type	:	Single-end suction, Self-priming centrifugal pump
3	Size (mm)	:	75 x 75
4	Overall Dimension and Weight (including mounting frame and engine)		
	4.1 Length (mm)	:	1250
	4.2 Width (mm)	:	490
	4.3 Height (mm)	:	22
	4.4 Weight (kg)	:	112
5	Impeller		
	5.1 Type	:	Enclosed
	5.2 Diameter (mm)	:	175
	5.3 Width (mm)	:	20
	5.4 No. of vane	:	8
	5.5 Material	:	Brass
6	Suction Inlet		
	6.1 Type	:	Threaded Flange
	6.2 Diameter (mm)	:	75
	6.3 Material	:	Aluminum
7	Discharge Outlet		
	7.1 Type	:	Threaded Flange
	7.2 Diameter (mm)	:	75
	7.3 Material	:	Aluminum

Table 1. (Cont'n)

11.	Casing	:	Volute
11.1	Type	:	Cast Iron
11.2	Material	:	Packing Type Seal
12.	Stuffing box	:	Counter clockwise when facing the suction inlet
13.	Rotation	:	
14.	Rated speed (rpm)	:	2400
15.	Rated Maximum Discharge Capacity (L/s)	:	No Data
16.	Rated Maximum Total Head (m)	:	No Data
17.	Primerover (based on Plate Rating)	:	
17.1	Brand	:	KATO
17.2	Model	:	KD80N PH
17.3	Serial No.	:	12090025
17.4	Make	:	No Data
17.5	Type	:	Four stroke cycle, water-cooled, single horizontal cylinder diesel engine
17.6	Rated Output Power (kW)	:	
	Maximum	:	6.25 @ 2400 rpm
	Continuous	:	No Data
17.8	Bore x Stroke (mm)	:	No Data
17.9	Displacement Volume (cc)	:	487
17.10	Cooling System	:	Water-cooled (thermo-siphon)
17.11	Lubricating System	:	Force-feed
17.12	Starting System	:	Manual cranking
17.13	Air Cleaner	:	Oil Bath
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in Table 2 and plotted in Figure 3.

Engine Tested	KATOK 10.3" x 3" coupled to KATOK 100000 Diesel Engine
Date of Test	April 1, 2014
Test Condition	
Ambient Temperature (°C)	34.3
Atmospheric Pressure (mmHg)	1013
Pumping Water Temperature (°C)	70.3

Table 2. Results of Performance Test

Test Point (mm)	Water Flow (mm)	Time taken, sec (sec)	Water Volume (mm)	Water Flow (mm)	Water Flow (mm)	Water Flow (mm)	Water Flow (mm)
2000	2433	1.76	13.0	0.00	0.00	13.1	0.00
2050	2368	1.94	13.8	12.7	0.03	26.3	14.8
2100	2343	2.24	12.2	16.1	3.48	33.3	13.0
2150	2324	2.44	10.6	17.3	3.34	35.3	14.0
2200	2303	2.41	10.4	18.8	3.40	35.1	13.0
2250	2223	2.34	10.3	18.6	3.17	34.4	13.0
2300	2182	2.38	10.0	20.3	2.03	33.7	12.3
2350	2183	2.36	10.0	20.0	2.03	33.3	11.3
2400	2152	2.31	11.0	21.3	2.30	33.0	10.3
2450	2100	2.30	9.13	20.0	1.67	34.3	7.03

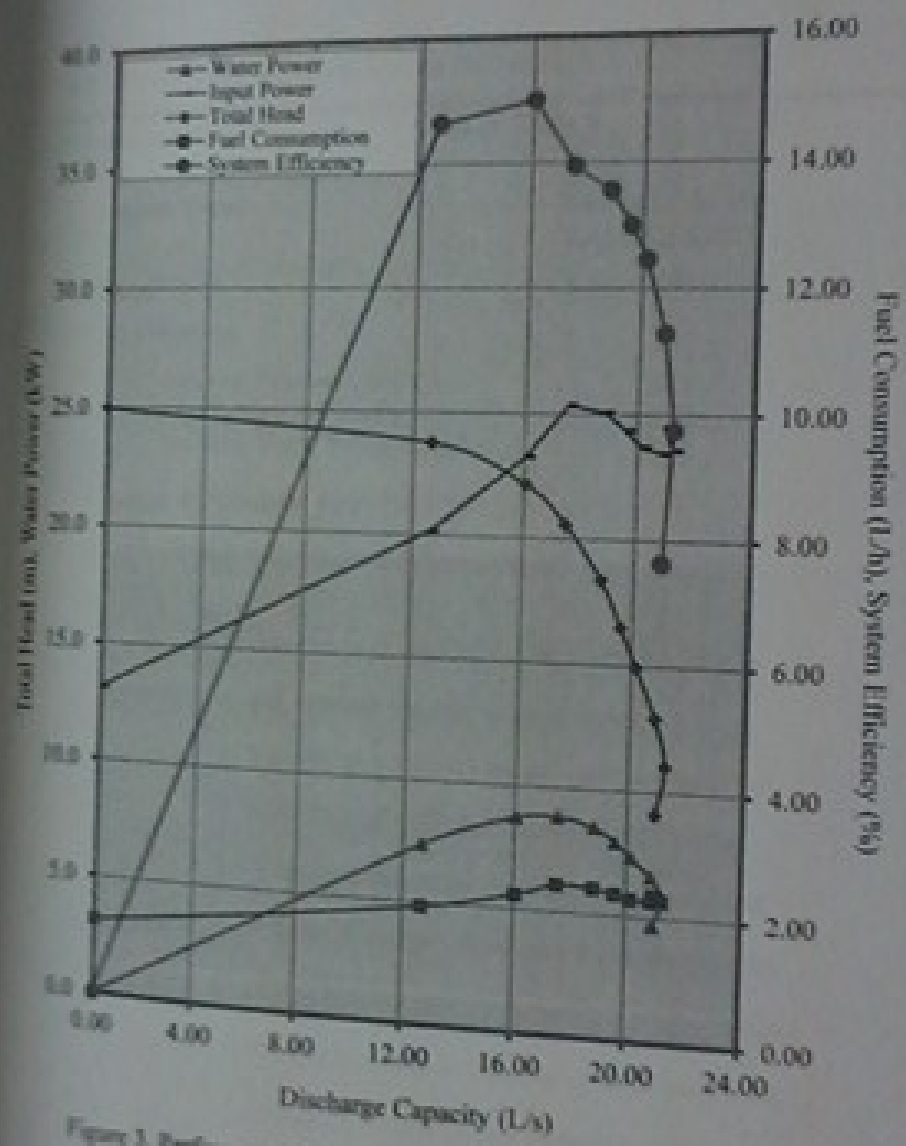


Figure 3. Performance characteristic curves of KATO KC30 3"X3" Centrifugal Pump coupled to KATO KD80N PH Diesel Engine

DATA COMPUTATION

ANNEX I. Formula used during calculation and testing

Total Head (m), H

$$TDH = H_d - H_s$$

Where: $H_d = \frac{P_d}{\gamma} + \frac{V_d^2}{2g} + Z_d$

$$H_s = \frac{P_s}{\gamma} + \frac{V_s^2}{2g} + Z_s$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_d}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_s}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_d^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_s^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Notes (Cont'd)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, $NPSHA$. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineers:

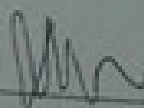

CONRAD IAN S. TAN


ROMULO E. EUSEBIO

Checked by:


DARWIN C. ARANGUREN

Approved for release:


DELFIN C. SUMINISTRADO
Director

MAY 12, 2014
Date

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